

Exam in Introduction to Machine Learning, 2DV516, 7.5 hp

August 16, 2024, 8.00–13.00

The exam consists of 10 questions. The maximum number of points for the exam is 40. In order to pass (grade E) you need 20 points.

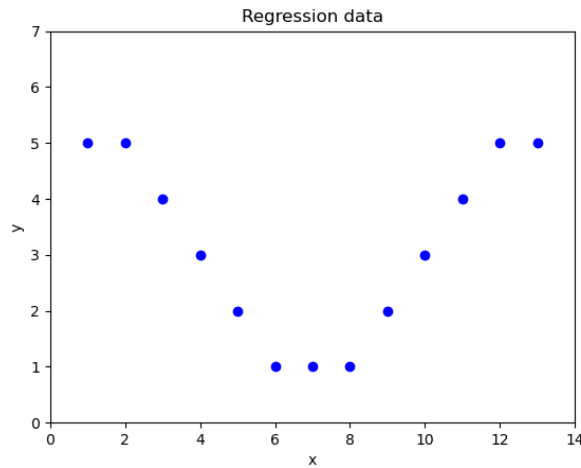
1. k-Nearest Neighbors (kNN)

Figure 1: Data sets for regression.

- (a) Present a detailed algorithm for how to classify an arbitrary data point x, y using kNN classification. You can assume binary classification data (0 or 1) and that distances between two points x^i, y^i and x^j, y^j and are measured using the Euclidean distance

$$d = \sqrt{(x^i - x^j)^2 + (y^i - y^j)^2}.$$

(3p)

- (b) Make a copy of the regression data above and sketch the regression plot for kNN with $k = 3$.

(3p)

2. Bias-variance tradeoff.

- (a) Explain how model complexity affects bias and variance. Discuss why certain models are considered low-complexity or high-complexity and how this influences their performance. Provide an example of a low-complexity model and a high-complexity model, discussing their characteristics and the impact on bias and variance. (2p)
- (b) Suppose you have trained two models on the same dataset: Model A is a simple linear regression model, and Model B is a complex polynomial regression model. Model A has a training error of 2 and a test error of 30. Model B has a training error of 10 and a test error of 15. Analyze the performance of both models in terms of bias and variance. Which model would you choose and why? (2p)

3. Linear and Logistic Regression.

- (a) Compare and contrast linear regression and logistic regression in terms of the types of problems they are designed to solve and the assumptions they make about the data. (1p)
- (b) In logistic regression, what is the purpose of the log-likelihood function? Describe how it is used in the training process. (2p)

4. Model selection and regularization.

- (a) Describe how you would use grid search in conjunction with cross-validation to select the optimal hyperparameters for a machine learning model. (2p)
- (b) In the context of Lasso Regression, explain what happens to the coefficients of irrelevant features as the regularization parameter (λ) increases. Why does this happen? (2p)
5. Figure 2 shows a neural network with two inputs, one hidden layer with two nodes and one output. All the activation functions $f_h(x)$ in the hidden units are rectified linear units (ReLU) defined as

$$f_h(x) = \max\{x, 0\}.$$

The output activation f_o is a step-function

$$f_o(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Give, if possible, examples of inputs $(x_1, x_2) \neq (0, 0)$ such that the output is 0 and 1 respectively. You only need to provide one example for each output (if it exists). (3p)
- (b) Write down the matrices corresponding to the transition between the input and hidden layer, and the hidden and output. (2p)

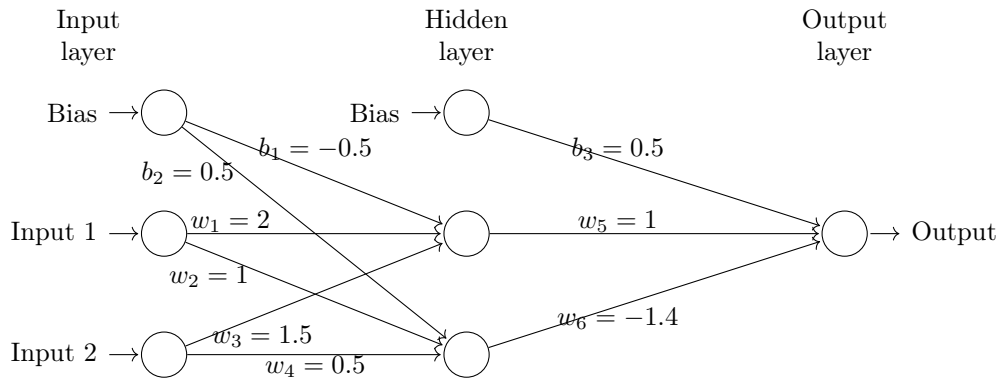


Figure 2: A neural network with one hidden layer consisting of two nodes.

6. The purity gain for a binary split in a decision tree is determined using the formula

$$\Delta = I(\text{parent}) - \sum_{j=1}^2 \frac{N(v_j)}{N} I(v_j),$$

where I denotes an impurity measure, $N(v_j)$ represents the number of elements in the node v_j (*i.e.*, the j th subregion after the split), and N is the total number of elements in the parent node (*i.e.*, the region where the split is made).

In this exercise, we use the Gini impurity as our impurity measure, defined for the m th region by

$$G_m = 1 - \sum_{k=1}^K \hat{p}_{mk}^2,$$

where $\hat{p}_{mk}$ is the proportion of training observations in the m th region that belong to the k th class.

Using the three-class data in Table 1, calculate the purity gain for both splits S_1 (split based on leaf count ≥ 6.5) and S_2 (split based on height ≥ 45). Which split is preferred based on your calculations? (5p)

height (cm)	30	40	50	35	55	45	60	50
leaf count	5	8	7	5	4	6	9	7
flower color	R	W	B	R	W	B	W	W
Class	P	Q	Q	P	R	P	R	Q

Table 1: A table of plant data including height, leaf count, and flower color (Red, White, Blue) with their classification (P, Q, R).

7. In Figure 3 we have a linearly separable binary classification problem. Consider this dataset when answering the following questions.

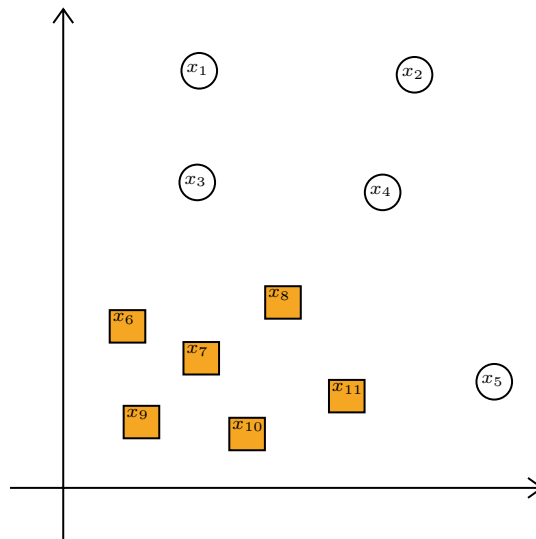


Figure 3: A linearly separable binary classification problem

- (a) Although this problem is linearly separable, argue why a quadratic kernel might be preferred. (2p)
- (b) With a quadratic kernel, which datapoints are likely to be support vectors? (1p)
8. Describe three differences between the clustering algorithms k -means and DBSCAN, indicating, for example, in which scenarios one works better than the other, or which characteristics one offers that the other does not. (3p)

9. Consider the pairwise *dissimilarity* matrix below, computed from four observations.

$$\begin{bmatrix} 0.0 & 0.3 & 0.4 & 0.7 \\ 0.3 & 0.0 & 0.5 & 0.8 \\ 0.4 & 0.5 & 0.0 & 0.45 \\ 0.7 & 0.8 & 0.45 & 0.0 \end{bmatrix}$$

- (a) Apply hierarchical clustering with complete linkage, and draw the resulting dendrogram. Indicate correctly the height at which the merges occur (but it does not need to be *perfectly* measured). (3p)
- (b) Extract two clusters and indicate which points are in each (consider point 0 as being the first row, point 1 as the second row, and so on). (1p)

10. Consider the example in Figure 4 of two different possible projections (i.e. dimensionality reduction) of a simple 2D dataset. Discuss the pros and cons of each projection, and why (in which scenario) you would use one over the other. (3p)

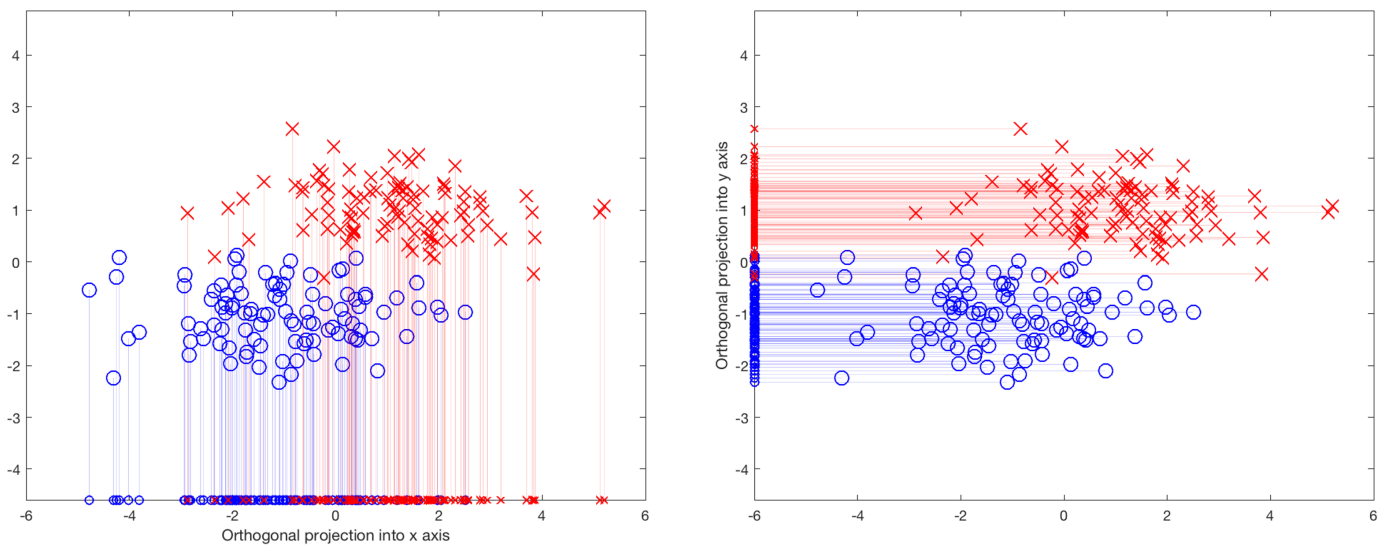


Figure 4: Two different projections of the same dataset (for Question 10).

Good Luck!